Personalizing CPAP Intervention Patterns with Digital Biomarkers and Patient Feedback

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***Abstract*— The continuous positive airway pressure device is effective in the treatment of obstructive sleep apnea, but real-world outcomes are limited by inconsistent Adherenec and generic interventions. The mismatch between interventions and premature disengagement often results from treating heterogeneous comorbidities or therapy stages as similar, leading to the same level of support. We propose an adaptive framework that provides interventions based on stage of therapy, comorbidity, and biomarker-derived risk. The framework integrates CPAP telemetry (usage, residual AHI, leak, and pressure), physiological data (HRV and oxygen desaturation), and patient-reported feedback to provide an update on the patient's state and prescribe the next-best interventions. Personalization is a learning process, whereby an improved comprehension, direction, and feedback, patients can transition to the Non-Adherent and Attempted category and the Attempted and Adherent category that allow the model to vary the intensity and content of the intervention. Mining ordered intervention sequences determines effective and ineffective patterns and triggers clinician review when impact is absent. Review of over 40,000 Linde HomeCare patients indicated that technical visits and escalation phone calls were better nightly utilization in early-stage users, and repeated contacts by SMS were neutral or negative linked to increased dropout. Initial results indicate that, this method reduce mistargeted outreach, improve self-efficacy and satisfaction, and positively affect multimorbid patients outside the one-size-fits-all workflow. The aim is a scalable personalization pipeline that can be used in homecare and balances algorithmic recommendations with physician supervision. In short, we measure short-term Δ-usage and durability (30/60/90-day ≥4 h; 6–12-month persistence) across risk groups via biomarker-informed personalization, supporting future clinical trials on Adherenec and quality of life.**

Keywords—CPAP Adherenec, personalization, digital biomarkers, Adaptive interventions, patient engagement

I. Introduction

Obstructive sleep apnea (OSA) is a widespread disease with major cardiovascular, metabolic, and neurocognitive consequences. Uncontrolled OSA predisposes to hypertension, arrhythmia, stroke, insulin resistance, and poor cognition in the day, and has an indirect impact on the quality of life and healthcare use[1]. The gold standard of therapy is continuous positive airway pressure (CPAP), yet real-world effectiveness remains limited by poor long-term Adherenec. In several cohort studies, despite initiation support, nearly one-third of patients end up not using or seriously not using CPAP in the first year[2]. This loss of engagement is not unique to poorly adherent starters: patients who appear to be complying according to device-defined compliance criteria sometimes drop out of therapy after months of apparent stable use, as seen in the so-called "adherent-but-leaving" phenomenon. The stratification currently used in clinical practice is mainly rule-based, relying on thresholds such as nightly use (≥4 h), residual apnoea-hypopnoea index, or mask leak to classify patients as adherent or Non-Adherent. Consequently, interventions are typically generic education, troubleshooting, or follow-up calls which fail to address patient diversity and can increase disengagement[3],[4]. Telemonitoring programs have increased the ability to detect usage lapses or technical issues, but systematic reviews show that even intensive monitoring alone does not sustain Adherenec improvements[5],[6]. Telemedicine-supported CPAP and protocolized education/augmented support suggest benefits when outreach is targeted and structured[7],[8]. OSA is heterogeneous in nature, patients are diverse both in terms of symptom profile, comorbidity burden, and pathophysiology. Recent cluster and endotype research suggest that loop gain, arousal threshold, and airway collapsibility are features that characterize different subgroups that react differently to therapy[9]. And engineering clinical updates emphasizing phenotype-aware algorithmic strategies[10]. Poor Adherenec trends are observed among multimorbid patients, particularly cardiovascular or metabolic disease patients, and these patients require support[11]. The early weeks of the therapy have a strong predictability of long-term persistence in terms of short-term trends in nightly use and mask leak[12]. Additional predictive value is provided by baseline questionnaires, such as the Epworth Sleepiness Scale and the Fatigue Severity Scale[13]. Besides these, digital biomarkers, such as heart rate variability and oxygen desaturation indices, offer physiologic sensitivity by sensing autonomic restoration and tolerance risk, which is not encompassed in a compliance measure[14],[15]. Existing

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CPAP interventions are not solely fixed but also predetermined and rule-oriented: once the Adherenec level is reached, the support plan is not likely to change, regardless of the patient's knowledge, comorbid condition, or emergence with different biomarkers. In practice, implies that not only is clinical or behavioral improvement of a patient not converted into new individualized behaviors, but also that the onset of early physiological or motivational decadence is not recognized until it is too late to intervene. To reduce this gap, personalization must connect patient comprehension with ongoing feedback and adaptive decision logic able to change the nature, timing, and magnitude of interventions. As the levels of understanding and personalization rise jointly, even the category of patients can change, Non-Adherent users will be able to transition to Attempted or adherent classes with quantifiable effects on the Adherenec outcomes and model development. This needs a reproducible, transparent system of personalizing interventions up to the level of choosing, sequencing, and modifying actions in real time depending on therapy stage, comorbidity profile, biomarker signals and patient feedback. In this work, we therefore investigate: (i) what are the strengths and weaknesses of current intervention strategies, (ii) how digital biomarkers can be used to further personalize treatment and predict the need to adapt, (iii) how comorbidity and stage-aware customization can be used to reduce dropout, and (iv) how feedback loops and participatory tools can be used to enhance patient engagement. Lastly, we discuss how such an adaptive model has an opportunity to provide quantifiable results of Adherenec, symptom management, daytime functioning, and quality of life in diverse groups of patients. Our clinical and technical goal is to extend the concept of intervention beyond static models by integrating multimodal data and patient-reported feedback into a risk-aware decision layer that alerts clinicians when automated assistance is inadequate. Through this adaptive personalization strategy, we will be able to establish the foundation of CPAP strategies that are clinically plausible, visible, and replicable in healthcare practices. We treat personalization as a learning process that recommends a next-best action, measures the 7-day response, and updates policy using ordered intervention sequences (motifs) as the unit of analysis.

II. METHODS

A. Personalization workflow and design pathway

The proposed framework provides a systematic personalisation process, which combines mix adaptive of educational, technical, motivational, and supportive interventions depending on the stage of the therapy, comorbidity profile, and dynamic digital biomarkers. Personalization goes beyond Adherenec limits that are maintained to an open-loop process where patient knowledge, physiological response and behavior reinforcement inform the next best action. The process is triggered at therapy initiation, when baseline telemetry and patient-reported data become available, and evolves through iterative reassessment as new intervention outcomes and biomarker trends are recorded. This framework has five operational steps. To form a baseline profile, first, device telemetry, physiological (heart rate variability, oxygen desaturation index) and patient questionnaires (e.g., ESS, FSS) are incorporated. Second, a rolling, 30-day Adherenec window is used to stratify patients into Adherent, Attempted, and Non-Adherent (Section B). Third, the system chooses and provides the most suitable intervention among the eight-category catalogue (educational, technical, behavioral, etc.) and mode (Visit, Call, or SMS) based on the risk profile of a patient. Fourth, response is measured using a symmetric 7-day Δusage window to measure short-term feedback loops that recalibrate the intensity of support. Lastly, an adaptive learning processes identify effective intervention sequences and motifs, simplifying subsequent decision rules. These steps move the workflow through an initial rule-based onboarding phase to a biomarker-driven adaptive phase during the first four to six weeks of therapy. This framework operationalizes the study aims: evaluating intervention strengths and weaknesses, using biomarkers for adaptive treatment, customizing by therapy stage and comorbidity, and employing feedback loops to enhance engagement. The subsections (B-I) below specify the sources of data, taxonomy of intervention, quantification measures, and analysis processes that execute this personalization pipeline in practice within a homecare setting.

B. Study population and data sources

We analyzed monitored Linde HomeCare records data for more than 40,000 patients treated with CPAP over a two-year observation window (13 March 2023 – 13 February 2025), stratified into three clinically meaningful cohorts that reflect persistence and usage behavior, namely (Adherent, Attempted, Non-Adherent) categories. On an individual basis, we have compiled longitudinal device telemetry (number of hours of nightly use, index of residual apnoea–hypopnoea, and mask leak), time-stamped intervention history, therapy status (continued vs. left therapy), and structured cancellation outcomes. The patient-reported outcomes of enriched subsets were Epworth Sleepiness Scale (ESS), Fatigue Severity Scale (FSS), and brief items of satisfaction/engagement. A priori inclusion and exclusion criteria served as valid identifiers, complete telemetry within the pre- and post-intervention period and at least three valid nights of data in a seven-day period. Outlier leak rate and negative values were eliminated. Adherenec classification was based on rolling 30-day windows, and intervention effects were calculated from symmetric 7-day pre-/post windows around each action. Patients with >14 consecutive nights of missing data (not due to hospitalization or equipment replacement) were excluded to maintain observability. “Left therapy” was defined by an explicit cancellation record or ≥30 days without device data confirmed by the provider log. Baseline demographics and comorbidity indicators were summarised as therapy persistence cohort level means and used to derive realistic gradients of therapy persistence. This dataset provided the foundation for analyzing attrition, intervention effectiveness, and persistence dynamics across categories.

C. Intervention taxonomy and delivery modes

The operational actions of the workflow in care were mapped onto an eight-category intervention catalogue: educational, technical, behavioral, motivational, supportive, escalation, structured follow-up, and administrative/scheduling. Interventions were delivered via three channels (Visit, Call, SMS), allowing for joint analysis of type and delivery mode. This two-axis (type × mode) model captures the strength, timing, and individualization of care interventions and links each contact to patient therapy stage and risk profile. Across all cohorts, over 40,000 actions were recorded in intervention logs. The less frequent but higher-impact contacts (≈10–15%) were in-person Visits, while Calls and SMS comprising ~78% of entries formed the high-volume

digital communication layer. Visits provided targeted, technical, and behavior-oriented support. Pattern analysis validated these differences between cohorts: Adherent patients were more likely than Attempted and Non-Adherent groups to get a higher proportion of educational and technical Visits indicative of routine optimization; Attempted and Non-Adherent groups had a larger proportion of escalation and motivational activity which was frequently precipitated by unstable usage or early decline. These variations help to explain the degree of adaptive personalization in diversification of intervention type and mode and the role of re-balancing between less intrtaction and high-intrcation efforts in facilitating category transition, i.e., from Attempted to Adherent trajectories.

D. Quantifying single-action impact on CPAP use

To estimate short-term effects, we computed within-patient changes in average nightly use across symmetric windows:

$$\Delta\, usage = \overline{\{usage_{\{+1\ldots+7\}}\}} - \overline{\{usage_{\{-7\ldots-1\}}\}} \qquad (1)$$

Interventions with incomplete windows were removed. Since deltas can be not only skewed but also stage-dependent, we give medians and interquartile ranges (IQRs), with means and sample sizes for completeness. The stratification is based on effects in cohort (Adherent, Attempted, Non-Adherent), and delivery method (Visit/Call/SMS). For stability, we required ≥5 actions per code×cohort cell. We ensured weekday/weekend balance within the pre/ post window and filtered the overlapping actions; when there was more than one action in the same window, we only kept the first action to isolate the effect of one intervention. This process produced approximately 38,000 valid pre/post windows across the entire dataset, with a median duration of approximately 6.2 h/night of baseline usage in adherent patients, 3.1 h/night in Attempted patients, and 1.2 h/night in Non-Adherent patients. The distribution checks were done to ensure that the estimates of change in usage were not systematically biased due to a seasonal effect or concentration of interventions around follow-up visits. Combined, these protection measures provided a strong platform in which to measure the effect of short-term intervention.

E. Therapy status and cancellation analysis

Persistence was reported in terms of the percentage of patients who were left in therapy at follow-up. Reasons of cancellation were then categorized into non-overlapping groups: hospitalization/death, intolerance/inobservance/no improvement, and physician request as ways of contextualizing the effects of interventions and detecting patterns where repeated low-touch behavior before intolerance-driven discontinuation occurred. Across all cohorts, ~15% of patients were flagged as Left therapy within 12 months, with markedly higher rates in the Non-Adherent group (~38%) than in Attempted (~21%) or Adherent (~7%). The highest number of discontinuations was through intolerance/no improvement, which made up over half of discontinuations. Elaborated records revealed that the high rate of SMS or generic follow-up calls was frequently concentrated within the 2-4 weeks preceding the intolerance to the cancellation and suggested that low-touch interventions were inadequate to stabilize high-sensitivity patients.

F. Patient-reported outcomes and digital biomarkers

Physiological sensitivity was also included with compliance through digital biomarkers of regular monitoring (heart rate variability [HRV] measures and nocturnal fragmentation and oxygen desaturation measures). Nightly biomarker series were summarized to nightly medians, winsorized at 1 st/99 th percentiles, and smoothed with 3-night moving median to reduce noise and, lastly, were standardized within the cohort. Biomarkers have been explored as baseline moderators of intervention response (interaction with Δusage) and dynamic triggers, i.e. whether changing physiology in otherwise compliant patients to the worse was a predictor of negative usage change and justified escalation.

G. Mining patterns of intervention sequences

To move beyond single actions, we reconstructed each patient's ordered intervention timeline and segmented it using ≥7-day gaps as a no-preemption buffer to limit overlap. For each contiguous sequence (maximum length three), we calculated its aggregate effect using two complementary definitions: (i) the sum of per-action Δusage within the sequence and (ii) the net post- vs. pre-sequence 7-day average. We then identified frequent sequences (occurring ≥10 times or in the top decile of sequence frequency within the cohort) and classified them as positive motifs when their median aggregate effect exceeded +0.2 h/night with the entire IQR above zero, or negative motifs when below −0.2 h/night with the IQR entirely below zero. Motifs were compared across cohorts and modes to expose stage-specific orchestration rules, e.g., whether early escalation (Call → Visit) converts negative single-action effects into net positive trajectories in unstable users, or whether repeated SMS contacts (SMS → SMS) frequently precede disengagement in high-risk groups. This approach enables identification of recurrent act-impact motifs that reflect personalization potential.

Across all cohorts, we reconstructed ≈ 12,000 valid sequences, with adherent patients showing the largest proportion of positive motifs (≈ 42 %), while Attempted and Non-Adherent groups exhibited a higher prevalence of negative motifs (> 30 %). The most common positive motif was a Technical → Motivational Visit in Attempted patients, yielding median gains of +1.2 h/night. Conversely, repeated SMS → SMS calls in Adherent patients often generated neutral or negative Δusage (median −0.3 h/night), a pattern also observed with single-action analysis of saturation effects. Most importantly, patient-reported outcomes and the dynamics of digital biomarkers were compared with the dynamics of motifs to test the hypothesis that physiological recovery (e.g., an improvement of HRV or the decrease of desaturations) was linked to positive motifs and the degradation of physiology was associated with the negative ones, leading to dropout. In summary, patterns of reproducible orchestration: technical-motivational visits have a measurable benefit when driven by escalation or combined with technical-motivational visits and repetitive low-touch interaction, saturation or weakening, is evident with repetition via this sequence-mining layer. The patterns are quantitative proof of how adaptive sequencing of interventions may push patients into better Adherenec groups, which agrees with the dynamic personalization model developed in this paper.

H. Metrics and statistical approach

The primary clinical outcomes were the median Δusage by intervention and by sequence, ≥4 h/night at 30/60/90

days; time-to-dropout (6–12 months), and the change in questionnaire-derived symptoms (ESS, FSS) where applicable were primary clinical outcomes. The main patient-centered measures were regarded to be satisfaction and engagement, whereas the system-level measures include the distribution and intensity of intervention types/modes, cohort-specific persistence, and relative contribution of high- vs. low-touch actions to Adherenec improvement. We performed non-parametric summaries (medians and IQRs) with clustered bootstrap confidence intervals (1,000 resamples), Hodges–Lehmann median differences for continuous comparisons, and χ^2 or Fisher's exact tests for categorical proportions (two-sided $\alpha = 0.05$). To overcome intra-patient dependency and avoid overestimation of precision, patient-level clustered bootstrap resampling on a patient level was used to address repeated interventions in a single individual. Missing PROMs (e.g., ESS or FSS not recorded during certain follow-ups) were treated by pairwise deletion with sensitivity analysis confirming that the omission did not cause bias in median estimates. Multiplicity across cells (e.g., intervention×stage) was controlled using Benjamini–Hochberg false discovery rate (FDR) adjustment.

The sensitivity analysis of the motif aggregation (sum vs. net definitions) and sequence-frequency threshold (>=10 or top decile) confirmed that the relative rankings of the effects and the consistency of the signs (positive vs. negative motifs) remained unchanged. Further validation was made on the bootstrap medians of the random 80 % subsamples to complete estimates of the full cohort with less than 0.1 h/ night deviation, hence internal reproducibility was supported. A combination of these statistical decisions ensures that short-term Δusage estimates, dropouts, and shifts on the questionnaire are represented by a patient-level variation, and not one or several extreme values, providing a robust quantitative backbone for interpreting personalization outcomes.

I. Reproducibility safeguards

All the analyses were pre-specified and carried out in a controlled analytic environment to guarantee transparency and independent verification. The workflow had a 7-day pre/post symmetric window, cohort stratification, and the minimal sample requirement with intervention sequences restricted to three actions and subject to a predefined frequency cut-off. Automatic filters on overlapping actions filtered these actions and sensitivity analyses were used to test the strength of the various aggregation definitions (sum versus net) and sequence-frequency thresholds. The records were de-identified and deterministically connected across telemetry, intervention, and questionnaire tables using patient-specific time stamps, allowing traceable and non-duplicated joins. These procedures guaranteed that Δusage estimates, motif classifications, and cohort summaries are reproducible and can be independently replicated on future data, in line with digital-health reproducibility standards.

III. Results and Discussion

The analysis of three large patient cohorts (~9,200 adherent, ~1,200 Attempted, and ~500 Non-Adherent) showed that there was a distinct difference between the persistence of therapy, with only 3.1% of patients who Adhered to therapy dropping out of therapy, and 6.0% of Attempted and 20.3% Non-Adherent, respectively. This gradient indicates that early instability is an effective predictor of disengagement, supporting stage-sensitive personalization over homogeneous strategies. The intervention catalogue consisted of eight categories offered through Visits, Calls, or SMS, although their distribution was highly skewed. The workflow was dominated by generic SMS and generic Calls, and other more specific or intensive tasks like motivational visits or escalation calls were underused. The result of this imbalance: in adherent patients, such generic high-volume interventions tended to result in neutral or negative effects on usage (median≈–0.2 h/night), indicating that repeated low-impact contacts may cause saturation or even disengagement. In comparison, the response rate in Attempted patients was higher, especially regarding technical visits (+1.6 h/night) and escalation calls (+0.7 h/night), which suggests that early-stage users are more intervention-sensitive when support is properly focused. The Non-Adherent responses were not consistent and only showed some minor and temporal corrections following escalation visits suggesting that interventions occurring after physiologic or behavioral events are usually late in order to alter the course. The reasons of cancellation were also investigated in all Adherent patients who stopped; over half did so because of hospitalization or death but a subgroup provided a reason of intolerability or nonbenefit, an instance of adherent-but-leaving phenomenon. In Attempted and Non-Adherent groups, intolerance, poor perceived benefit, and disengagement were the most prevalent dropout causes, aligning with negative Δusage following repetitive SMS. Across 30/60/90 days, positive sequences maintained higher ≥4 h/night and lower 30-day dropout, and early positive sequences predicted greater 6–12-month persistence. The assistance of patient-reported outcomes and digital biomarkers was beneficial to get further knowledge. These physiologic predictors including heart rate variability and oxygen desaturation indices served as dynamic moderators: positive usage sequence tended to be preceded by improvements in biomarker profiles whereas negative motifs and following dropout were frequently preceded by deteriorations. It is therefore possible to provide a pathway to these physiological signals that can activate timely escalation before the behavioral decline is established. Intervention sequences showed repeatable act impact patterns. Among Attempted patients, early intensification (Call → Visit) frequently converted negative single-action changes into net gains, supporting timely escalation. The use of repeated SMS in Non-Adherent patients without escalation had often resulted in discontinuation, which underlines the risks of over-utilization of generic low-touch measures in high-risk populations. Adherent patients were not spared in this: the regular contacts of frequent routine, when not tailored, sometimes led to overall declines in its application with overall stability. In a temporal split, effects and motif ranks were acquired on a posterior period extrapolated to a posterior nonoverlapping (temporally) cohort with retained direction and rank order (calibrated next-best-action deciles). The results of a longitudinal analysis indicate that the results of participants who were exposed to positive motifs experienced significantly lower dropout rates within 30 days than did participants who were exposed to negative motifs, thus demonstrating the influence of sequence composition and not isolated acts in determining the effects. Collectively, these results indicate that CPAP support cannot be based on static and generic interventions. Various patients at various points of therapy react differently, and repetition of low-touch behaviors can lead to worse Adherenec. The advantages of patient-reported outcomes and digital biomarkers

incorporated in the escalation rules provide an opportunity for earlier, more precise escalation. Although exploratory, these findings give empirical support to the fact that adaptive personalization is possible: expanding the intervention catalogue, acting on stage and comorbidity profile, and triggers by using physiologic signals. The advantages are founded on the observational history of homecare attendance that may be confounded with indication and limited causal inference, and absence of biomarker and questionnaire data in each cohort. Also, short symmetric 7-day windows were analysed regarding the effects of the intervention, and long-term sustainability is yet to be evaluated. Nonetheless, the consistency of positive and negative intervention motifs and their relation to patient-reported and physiological cues give the basis to a personalization framework that ensures a balance between the coordination of algorithms and the supervision of a clinician.

IV. Conclusion

This research shows that CPAP support is not a fixed process, but rather a dynamic behavior across stages of therapy, patient factors, and the ordered sequence of interventions. This paper demonstrates, at scale, personalized treatment of OSA through dynamic personalization based on patient stratification, multimodal data, and sequence-aware orchestration of interventions into a coherent adaptive response by using over 40,000 real-world records of home-care. We discovered that early and unstable users were very responsive to specific technical visits and escalation calls, and recurrent low-contact interventions, including SMS, were neutral or even detrimental, especially in Non-Adherent groups. This proves that effective personalization requires not only identifying who is at risk but also adjusting the mode, timing, and intensity of support to the patient's changing state. Adherent patients can become disengaged even when being supported in ways that are not dynamic. Moreover, biomarker and questionnaire data were not completely covered, and thus, there was limited comprehensive coverage of the physiological and experiential aspects of personalization. The effects of intervention were assessed at short symmetric windows, and their sustainability in the long run is yet to be determined. Nevertheless, the outcomes provide good empirical data in favour of dynamic and open personalization without fixed limits. In this manner, the disparity between data-driven stratification and clinical decision-making can be reduced through a repeatable approach to the adaptive, patient-specific planning. The prospective studies will be used in future to verify motifs based on enhanced digital biomarkers and incorporate adaptive frameworks into the daily services under the supervision of clinicians. This model can emphasize real-time decision support, continuously adapting as physiological and behavioral patterns evolve. The potential improvements of CPAP Adherenec and patient quality of life, as well as the healthcare utilization and cost reduction, are not only tangible clinical and system-level values but also potential outcomes of such advances.

Acknowledgment

This paper presents results that are developed in collaboration between the Linde HomeCare France company and the University Lumiere Lyon 2, DISP Lab. This research is established under a CIFRE contract (2023/1249). The content of this paper reflects an R&D initiative promoted by Linde Homeare France. Responsibility for the information and views expressed in this paper lies entirely with the authors.

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